FIN2020: Foundation of Finance

Spring 2025

Instructor

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Lecture Location & Time:

L01 & L02:

TC308

Mon/Wed 8:30-9:50 AM, 10:30-11:50 AM

L03:

TD102

Mon/Wed 13:30-14:50PM

Tutorial Location & Time:

TBA

Note: *Tutorials are not available every week,* detailed times will be announced in advance

Course Description

Foundation of Finance is an introduction to the foundations of modern financial economics. The focus throughout will be on the development and interpretation of discrete-time models of asset pricing, and the valuation of risky cash flow. After developing and studying the details of consumer decision-making under uncertainty, it uses that general framework as a basis for understanding both equilibrium and no-arbitrage theories of securities pricing, including traditional models like the capital asset pricing model (CAPM) and the arbitrage pricing theory (APT), newer Arrow-Debreu theories and, if time permits, martingale pricing methods. The course is primarily theoretical. However, I may discuss some empirical puzzles in finance.

This course is intended for second year undergraduate students in finance and economics. While there are no formal prerequisites for enrolling in this course, a working knowledge of **calculus**, linear algebra, and **probability plus statistics** is especially useful.

Homework and Tests

The course grade will be based on the following: Class participation in the form of regular homework assignments (20%); a midterm exam (35%); and a final exam (45%). Homework will be assigned about every week and will usually be due one week after it is assigned. However, if you have a time conflict, feel free to drop us a line. NO score for homework turned in after the deadline. Students are encouraged to discuss the homework problems with other students, but should work through all of the mathematics individually, and all students must turn in separate assignments. **If we find two or more identical answers, then all students involved are subject to disciplinary actions.** You can also stop by during my or TA's office hours. I prefer to talk to you in person. I feel that Email is not a very efficient way to ask questions (hard to type equations).

As for the midterm and the final exam. There will be no make-up exams, and any conflicts or emergencies should be approved by me in advance of the exams. In case of illness or accident at the time of the midterm – with proper documentation from a doctor or the police – the final will be weighted 80%.

Grading Policy

If you think a mistake was made in grading your exam, you may ask for a regrade. You must request for regrading and turn in your exam within one week of when the exam is first returned to the class. If you write in pencil, however, you can pick up the exam from my office, check the grading immediately, and take care of complaints before leaving the office. If you ask for a regrade, your whole exam is subject to regrading. This may bring to light some previously unnoticed errors, and you may end up with a lower score, not higher. Note that unless your answer is fully correct, the assignment of partial

credit is a matter of judgment, and we are unlikely to change your grade since we want to treat all class members fairly.

Textbooks

The main textbooks corresponding to the suggested reading are given below. **The textbooks are NOT required. Some people find the textbooks helpful and some people do not.** Read the description below before you decide on whether you feel it would be worth it for you to buy the textbooks:

(1) *Intermediate Financial Theory* by Jean-Pierre Danthine and John B. Donaldson, 2014 (Undergrad Level or above)
(2) *Asset Pricing* by John H. Cochrane, 2005 (Master Level or above)
(3) *Asset Pricing and Portfolio Choice Theory* by Kerry Back, 2010 (PhD Level)

Additional textbooks are also helpful: *Theory of Financial Decision Making* by J. Ingersoll, 1987, and *Foundations for financial economics* by C. F. Huang and R. H. Litzenberger, 1988. I will refer to these texts as (1) DD, (2) JC, (3) KB, JI, and HL, respectively.

Below is a list of topics to be covered. I reserve the right to change the syllabus (modestly) as the course progresses. Under each topic I summarize relevant reading and provide a number of classic references. Students are recommended to read the textbook readings marked with *. Students who plan to apply for graduate programs in finance and economics are encouraged to read the book materials without *. Students who are serious about research and plan to apply PhD programs should also read some additional articles without *. I will generally provide lecture slides or notes summarizing key points made in class.

Course Materials

Lecture slides are THE main reference for your study. I will post lecture slides, some class readings, and homework assignments on https://bb.cuhk.edu.cn The lecture slides are based on Prof. Peter Ireland's lecture notes. Thus, the copyright of the slides belongs to Peter Ireland. Redistribution is permitted for education and research purpose, so long as no charges are made. All copies must be provided free of charge and must include the copyright notice.

Honor Code

All students are expected to adhere to the University's *Code of Academic Integrity*, which is designed to ensure that the principles of academic honesty and integrity are upheld. The Smith School does not tolerate academic dishonesty. All acts of academic dishonesty will be dealt with in accordance with

the provisions of this code. Please visit the following website for more information on the University's Code of Academic Integrity: http://www.cuhk.edu.cn/departsite/ar/en/Academic.html

Policies on Absence

Absences from class because of illness or important schedule conflicts are sometimes inevitable and should be fine so long as they are isolated in number. If you can't make it to class on any given day, you can check the course webpage to see the slides you missed. If you miss class on a day when a problem set is due, please scan your answers and email them to TAs before the deadline.

Special Needs

Any student with special needs should bring this to the attention of the instructor as soon as possible, but not later than the second week of class.

Course Outline

A. Introduction (2 weeks)

- 1. Overview of Asset Pricing Theory
- 2. Mathematical and Economic Foundation

Readings:

- *DD chapter 1,2
- KB chapter 1.
- Machina, Mark, 1982, `Expected utility analysis without the independent axiom.' Econometrica 50, 277-323.
- Machina, Mark, `Choice under Uncertainty: Problems Solved and Unsolved' Journal of Economic Perspectives, vol. 1, no. 1, Summer 1987, pp. 121-54
- Rabin, Matthew, `Risk Aversion and Expected-Utility Theory: A Calibration Theorem' Econometrica, vol. 68, no. 5, September 2000, pp. 1281-92.

B. Decision-Making Under Uncertainty (2.5 weeks)

- 3. Making Choices in Risky Situations
- 4. Measuring Risk and Risk Aversion

Readings:

- *DD chapter 3,4
- J. Pratt, 1964, `Risk aversion in the small and in the large.' Econometrica 32, 122-136.
- HL Chapter 2
- Ali, Mukhatar, `Stochastic Dominance and Portfolio Analysis', Journal of Financial Economics, 1975, Vol. 2, pp. 205-229.
- Rothschild, Michael, and Joseph Stiglitz, `Increasing Risk I: A Definition', Journal of Economic Theory, 1970, Vol 2, pp. 225-43.
- Levy, Haim, `The Definition of Risk: An Extension,' Journal of Economic Theory, 1977, Vol. 14, 232-34.

C. The Demand for Financial Assets (2.5 weeks)

- ❖ 5. Risk Aversion and Investment Decisions
- ❖ 6. Modern Portfolio Theory

Readings:

- *DD chapter 5,6
- JI chapter 2.
- New Palgrave, Finance, `Arbitrage?', Macmillan Press, New York, 1987, pp. 57-71.
- Ross, Stephen, 1978, `A Simple Approach to the Valuation of Risky Streams? Journal of Business, 50, 453-75.
- Dybvig, Philip H; Ross, Stephen A, `Portfolio Efficient Sets?, Econometrica, vol. 50, no. 6, November 1982, pp. 1525-46
- Fishburn, Peter, Burr Porter, `Optimal Portfolios with One Safe and One Risky Asset: Effects of Change in the Rate of Return and Risk?, Management Science, 1976, Vol 22, p. 1064-73.
- Samuelson, Paul A., `General Proof that Diversification Pays?, Journal of Financial and Quantitative Analysis, 1967, vol. 2, pp. 1-13.

Midterm: ***Note-Midterm may happen either before or after CAPM model mentioned below***

D. Classic Asset Pricing Models (3 weeks)

- ❖ 7. The Capital Asset Pricing Model
- ❖ 8. Arbitrage Pricing Theory

Readings:

- *DD chapter 8,14

- KB Chapter 5,6
- Merton, R., 1972, `An analytical derivation of the efficient portfolio frontier.' Journal of Financial and Quantitative Analysis,7, 1851-1872.
- G. Chamberlain, 1982, `A characterization of the distributions that imply mean-variance Utility functions.' Journal of Economic Theory 29, 185-201.
- Markowitz, Harry, "Portfolio Selection," *Journal of Finance* 7:77-91, 1952.
- Ross, S., "Mutual Fund Separation in Financial Theory—the Separating Distributions," *Journal of Economic Theory*, 17:254-286, 1978.
- W. Sharpe, 1964, `Capital asset prices: A theory of capital market equilibrium under conditions of risk.' Journal of Finance 19, 425-442.
- JI chapter 7.
- Huberman, Gur, "A Simple Approach to Arbitrage Pricing Theory," *Journal of Economic Theory* 28:183-191, 1982.
- Chamberlain, Gary, "Funds, Factors, and Diversification in Arbitrage Pricing Models," *Econometrica* 51:1281-1304, 1983.
- S. Ross, 1976, `Arbitrage theory of capital asset pricing.' Journal of Economic Theory 13, 341-360.

E. Arrow-Debreu Pricing (2.5 weeks)

- ❖ 9. Arrow-Debreu Pricing: Equilibrium
- ❖ 10. Arrow-Debreu Pricing: No-Arbitrage

Readings:

- *DD Chapter 9,11
- KB Chapter 2,3
- Hirshleifer, J., `Investment decisions under uncertainty: Application of state-preference Approach.' Quarterly Journal of Economics, May 1966, 252-277.
- Myers, S., 1968, `A time-state preference model of security valuation.' Journal of Financial and Quantitative Analysis, 1-33.
- Case, D., and J. Stiglitz, 1970, `The structure of investor preferences and asset returns, and separability in portfolio allocation: A contribution to the pure theory of mutual funds.' Journal of Economic Theory 2, 122-160.

F. Extensions (2.5 weeks)

- ❖ 11. Martingale Pricing *(Note: Depending on progress, we may not cover the entire chapter)*

Readings:

- *DD Chapter 10,12
- KB Chapter 4
- J. Lucas, Robert E. *Asset pricing in an exchange economy.* Econometrica, 46(6): 1429-1445, 1978.
- Rubinstein, M., 1976, `The valuation of uncertain income streams and the pricing of options., Bell Journal of Economics 7, 407-425.
- Black, F. and M. Scholes, *The pricing of options and corporate liabilities.* Journal of Political Economy, 81(3): 637-654, 1973.
- S. Ross, 1976, `Options and efficiency.' Quarterly Journal of Economics 90, 75-89.

Final Exam: TBD

This schedule is tentative. Readings might be revised as we go through the material.